Due: April 22nd, before class.

Name: Solution

Collaborators: N/A

Document your work and note any collaborators. In your write-up, please neatly document any code you have written, alongside its output, and including any necessary commentary to answer the question. Please use an `.ipynb` notebook for any code, include commentary in markdown code blocks, and print the `.ipynb` notebook to PDF to attach to your submission. The LaTeXsource for this file may be helpful: https://www.overleaf.com/read/nqxvfxsszypy#027369.

Please make sure to label the parts of your assignment! Use markdown blocks in your Julia notebook and make headings using #, ##, and ###.

Problems:

1. One way to approximate the eigenvalue problem $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ over $\mathcal{K}_m$ is the restrict $\mathbf{x}$ to the low-dimensional spaces $\mathcal{K}_m$.

   (a) Starting from $\mathbf{A}\mathbf{Q}_m = \mathbf{Q}_{m+1}\mathbf{H}_m$, show

   $$\mathbf{Q}_m^\top \mathbf{A}\mathbf{Q}_m = \tilde{\mathbf{H}}_m,$$

   where $\tilde{\mathbf{H}}_m$ is the upper Hessenberg matrix resulting from deleting the last row of $\mathbf{H}_m$. What is the size of this matrix?

   We can multiply both sides by $\mathbf{Q}_m^T$ to get

   $$\mathbf{Q}_m^T \mathbf{A}\mathbf{Q}_m = \mathbf{Q}_m^T \mathbf{Q}_{m+1}\mathbf{H}_m.$$

   Since $\mathbf{Q}_m$ and $\mathbf{Q}_{m+1}$ are orthonormal, their product would be a $m \times (m+1)$ matrix $\begin{bmatrix}\mathbf{I} & \mathbf{0}\end{bmatrix}$. Thus when this matrix is multiplied with $\mathbf{H}_m$ we would get a $m \times m$ matrix $\mathbf{H}_m$ with last row removed, which is $\tilde{\mathbf{H}}_m$.

   (b) Show the result above leads to the approximate eigenvalue problem $\tilde{\mathbf{H}}_m\mathbf{z} \approx \lambda\mathbf{z}$. (Hint: Start with $\mathbf{A}\mathbf{x} \approx \lambda\mathbf{x}$, and let $\mathbf{x} = \mathbf{Q}_m\mathbf{z}$ before applying part (a).

   Plugging in $\mathbf{x} = \mathbf{Q}_m\mathbf{z}$, we get

   $$\mathbf{A}\mathbf{Q}_m\mathbf{z} \approx \lambda\mathbf{Q}_m\mathbf{z}.$$

   Left multiplying both sides by $\mathbf{Q}_m^T$ gives

   $$\mathbf{Q}_m^T\mathbf{A}\mathbf{Q}_m\mathbf{z} \approx \lambda\mathbf{Q}_m^T\mathbf{Q}_m\mathbf{z}.$$

Since $\mathbf{Q}_m$ is orthonormal, $\mathbf{Q}_m^T\mathbf{Q}_m = \mathbf{I}$. Also from (a) we know that $\mathbf{Q}_m^T\mathbf{A}\mathbf{Q}_m = \tilde{\mathbf{H}}_m$, so we get

$$\tilde{\mathbf{H}}_m\mathbf{z} \approx \lambda\mathbf{z}.$$

(c) Apply the Arnoldi iteration method (use and cite code from class) to a matrix generated as

```
.        λ = @.  10 + (1:100)
.        A = triu(rand(100,100),1) + diagm(λ)
```

using a random seed vector. Compute eigenvalues of $\tilde{\mathbf{H}}_m$ for $m = 1, \cdots, 40$, keeping track in each case of the error between the largest of those values (in magnitude) and the largest eigenvalue of $\mathbf{A}$. Make a log-linear graph of the error as a function of m.

2. In this exercise, you will see the strong effect the eigenvalues of the matrix may have on GMRES convergence. Let $\mathbf{B}$ be a diagonal matrix with the values $1, 2, \cdots, 100$ on the diagonal, let $\mathbf{I}$ be a 100×100 identity, and let $\mathbf{Z}$ be a 100×100 matrix of zeros. Also, let $\mathbf{b}$ be a 200×1 vector of ones. You will use `IterativeSolvers.gmres` with restarts, as in Lecture 24.

(a) Let $\mathbf{A} = \begin{bmatrix} \mathbf{B} & \mathbf{I} \\ \mathbf{Z} & \mathbf{B} \end{bmatrix}$. What are its eigenvalues? (Don't use a computer here!)

We know that $\mathbf{A}$ is upper triangular, so its eigenvalues are its diagonal entries. Thus its eigenvalues are the diagonal entries of $\mathbf{B}$ with multiplicity 2.

(b) Apply `gmres` with tolerance 10^{-10} for 100 iterations without restarts, and plot the residual convergence. You can use and cite the `gmres` code from class.

(c) Repeat part (b) with restarts every 20 iterations.

(d) Now, let $\mathbf{A} = \begin{bmatrix} \mathbf{B} & \mathbf{I} \\ \mathbf{Z} & -\mathbf{B} \end{bmatrix}$. What are its eigenvalues? (Don't use a computer here!)

Same logic as (a). But instead of the diagonal entries of $\mathbf{B}$ we get $\pm$ the diagonal entires of $\mathbf{B}$ with multiplicity 1.

(e) Repeat part (b). Which matrix is more difficult for `gmres`?